ViewGen

AI Viewport Generator for Unreal Engine

Plugin Documentation v2.2.0

Comprehensive guide for users, artists, and developers working with the ViewGen AI Viewport Generator plugin for Unreal Engine.

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Engine: Unreal Engine 5.7

Platforms: Windows, Linux, macOS

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1. Introduction and Overview

ViewGen is an editor plugin for Unreal Engine that bridges your 3D viewport with AI image and video generation. It captures the editor viewport, extracts depth maps and camera metadata, and sends this data to generation backends (ComfyUI, Gemini, Kling, and more) to produce AI-generated imagery that matches your scene's composition, lighting, and camera angle.

The plugin includes a full node-based graph editor for building custom ComfyUI workflows, support for LoRA models, a result gallery with history, and a 3D model pipeline that can convert generated images into importable static meshes via the Meshy API.

Key Features

- **Viewport Capture** — Captures the active editor viewport at native resolution with full camera metadata (FOV, focal length, position, rotation).
- **Depth Map Extraction** — Renders a linearized depth pass from the viewport camera for use as a ControlNet guide.
- **Multiple Generation Modes** — img2img, depth+prompt, text-only, Gemini, and Kling pipelines with per-mode parameter controls.
- **Node Graph Editor** — Interactive drag-and-drop graph editor supporting all ComfyUI node types plus custom UE source nodes. Includes V3 dynamic nested input parsing for newer nodes (ByteDance Seedance 2.0, etc.).
- **Drag-to-Graph** — Drag result thumbnails from the gallery directly onto the graph editor to automatically create LoadImage or LoadVideo nodes with file paths and preview thumbnails preloaded.
- **Video Generation** — Image-to-video via Kling Video, Wan I2V, and Google Veo3.
- **3D Model Pipeline** — Convert generated images to 3D meshes via Meshy, import as UE assets with collision and Nanite support.
- **GLB Texture Pipeline** — Extract, process, and repack textures on GLB models with optional xatlas UV repack and bilinear upres (2x/4x).
- **LoRA Support** — Add, remove, and weight-adjust LoRA models for fine-tuned generation styles.
- **Result Gallery** — Browse, select, and navigate generation history with thumbnail previews.
- **Workflow Save/Load** — Save and load graph editor workflows as JSON files, including ComfyUI template import.
- **Cost Estimation** — Shows estimated API costs in the status bar before generation starts, with user-configurable pricing tables.

- **Session Persistence** — Viewport captures, depth maps, generation results, graph thumbnails, and LoadImage node previews all persist across editor restarts. Cached data loads from disk without needing a ComfyUI connection.
- **3D Node Type Compatibility** — The graph editor recognises ComfyUI's 3D file type family (FILE_3D, FILE_3D_GLB, FILE_3D_OBJ, etc.) and comma-separated multi-type inputs, enabling proper connections between 3D generation and save nodes (e.g., Tripo → SaveGLB).

2. Installation and Setup

Plugin Installation

Place the `ViewGen` folder into your project's `Plugins/` directory. Regenerate project files, recompile the module, and enable the plugin via Project Settings. The plugin registers a toolbar button and a Window menu entry labeled "ViewGen".

External Requirements

ViewGen communicates with a ComfyUI server for most generation pipelines. The server must be running and accessible at the configured URL (default: `http://127.0.0.1:8188`). Certain generation modes require specific ComfyUI custom nodes to be installed.

Generation Mode	Required ComfyUI Custom Nodes
img2img / Depth+Prompt	ControlNet nodes, KSampler (built-in)
Gemini (Nano Banana 2)	NanoBanana2AIO custom node package
Kling Image 3.0	KlingImageGenerationNode
Flux Pipeline	x-flux-comfyui custom node package
Video (Kling/Wan/Veo3)	KlingImage2VideoNode, WanImageToVideoApi, Veo3VideoGenerationNode
Meshy 3D	MeshyImageToModelNode

API Keys

Some features require API keys configured in the ViewGen panel's **Connection** tab:

ComfyUI API Key (optional)

Required for paid API nodes such as **Nano Banana 2** (GeminiNanoBanana2), **Kling Image** (KlingImageGenerationNode), and **Kling Video** (KlingImage2VideoNode). These are ComfyUI

API nodes — they are *not* traditional custom nodes and will not appear in ComfyUI Manager. They are server-side nodes that become available when your ComfyUI instance is connected to the platform with a valid API key. If you only use local ComfyUI nodes (Stable Diffusion, ControlNet, etc.), you do not need this key.

How to get your ComfyUI API key:

1. Go to platform.comfy.org and create an account (or sign in).
2. Verify your email address.
3. Navigate to platform.comfy.org/profile/api-keys.
4. Click **Create API Key** and copy the generated key. Format: `comfyui-xxxxx`
5. Paste the key into the **ComfyUI API Key** field in ViewGen's Connection tab.

Note: Paid API nodes require credits on your ComfyUI account. You can purchase credits at platform.comfy.org under your account settings.

Meshy API Key

Required for Image-to-3D and Text-to-3D model generation via the Meshy service.

How to get your Meshy API key:

1. Go to meshy.ai and create an account (or sign in).
2. Navigate to your account settings and locate the API section.
3. Generate a new API key. Format: `msy-XXXX`
4. Paste the key into the **Meshy API Key** field in ViewGen's Connection tab.

3. User Interface Overview

The ViewGen panel is a nomad tab accessible via the toolbar button or Window > ViewGen. The interface is divided into a top section (viewport capture and results) and a bottom section with three tabs for different workflows.

Layout Structure

The panel is split into a top section and bottom section. The top section contains three columns: input thumbnails (left), main preview (center), and result gallery (right). The bottom section contains a three-tab system: **Basic**, **Graph Editor**, and **Connection**.

Toolbar

The toolbar at the top provides the primary action: **Capture Viewport** captures the active editor viewport and depth map simultaneously, populating the thumbnails.

Tab System

Below the viewport/results area, three tabs organize the plugin's controls:

- **Basic** — Artist-friendly controls: prompt, generation mode, checkpoint and LoRA selection, resolution, steps, seed, and the Generate button. Mode-specific settings (Gemini, Kling) appear conditionally based on the selected mode. An Advanced collapsible section provides access to depth/ControlNet, hi-res fix, defaults, and presets.
- **Graph Editor** — Full visual node-graph editor for building and editing ComfyUI workflows. Includes a horizontal splitter between the graph canvas and a node details panel, plus a bottom toolbar with Generate from Graph, Load, Save, Fullscreen, and Clear buttons.
- **Connection** — Server and API configuration: ComfyUI URL, API keys (ComfyUI, Gemini, Kling, Meshy), timeout settings, and model paths. Includes Test Connection and Refresh Models buttons.

Thumbnail Panel

The left column displays thumbnails for the most recent viewport capture and depth map. These update when you click Capture Viewport. The thumbnails reflect the viewport's native aspect ratio.

Preview Panel

The center area shows the most recently generated image at full size with a progress bar during generation. Navigate between results using the gallery or arrow controls.

Result Gallery

The right column displays a scrollable list of all generated images in reverse chronological order. The currently selected result appears at full brightness with a gold border, while unselected results are dimmed. Click any thumbnail to display it in the preview panel.

Status Bar

The bottom of the panel shows the current status (Ready, Capturing, Generating, etc.) and a progress bar with percentage during active generation.

4. Capturing the Viewport

Clicking **Capture Viewport** reads the active level editor viewport's backbuffer at native resolution and simultaneously renders a depth map from the same camera position. Both are displayed as thumbnails and stored for use during generation.

Camera Metadata

The capture also extracts camera metadata from the viewport client, including horizontal FOV, world position, rotation, and a calculated 35mm-equivalent focal length. When **Auto Camera Prompt** is enabled in settings, this metadata is automatically converted into a natural-language description and injected into your prompt.

Example auto-generated descriptions: "shot with wide-angle 24mm lens, high-angle shot looking down, moderate perspective, widescreen 16:9 framing."

Depth Map

The depth pass uses a temporary SceneCaptureComponent2D positioned at the viewport camera. It captures scene depth and linearizes it to a grayscale image where white represents close objects and black represents distant objects. The **Max Depth Distance** setting (default 50,000 Unreal Units) controls the far boundary of the depth mapping.

Capture Reuse

During workflow execution, if a viewport or depth capture already exists from a previous Capture Viewport click, the plugin reuses it instead of re-capturing. For depth maps, a re-capture is triggered only if the workflow's max_depth setting differs from the last capture. This allows you to capture once and generate multiple times without the viewport needing to be visible.

5. Generation Modes

ViewGen supports five generation modes, selectable via the Generation Mode setting. Each mode uses a different combination of viewport data, depth maps, and prompts.

Mode	Inputs	Best For
img2img	Viewport image + Prompt + optional Depth	Refining viewport composition with AI style
Depth + Prompt	Depth map + Prompt (no viewport image)	Preserving structure while changing visuals entirely
Prompt Only	Text prompt only	Pure creative generation without viewport reference
Gemini	Optional viewport/depth + Prompt	Gemini-powered generation with flexible inputs
Kling Image 3.0	Optional viewport + Prompt	High-quality generation with subject/face fidelity

Per-Mode Parameters

Each generation mode exposes its own relevant parameters directly in the Basic tab. For ComfyUI modes (img2img, Depth+Prompt, Prompt Only), you control denoising strength, CFG scale, steps, and seed individually. For Gemini, you can adjust the model variant, aspect ratio, resolution tier, and thinking level. For Kling, you configure the model version, aspect ratio, image count, and image fidelity. All parameters are accessible without switching tabs.

6. The Graph Editor

The graph editor lives in its own dedicated **Graph Editor** tab, providing a fully interactive node-based workflow builder. It supports all ComfyUI node types (discovered dynamically from the server) plus custom UE source nodes for viewport capture, depth, camera data, and more. The tab includes a horizontal splitter between the graph canvas and a node details panel, with a toolbar pinned to the bottom.

Navigation

- **Pan:** Right-click drag or middle-mouse drag to move the canvas.
- **Zoom:** Scroll wheel to zoom in/out, centered on the cursor.
- **Fullscreen:** Click the Fullscreen button to open the graph in a dedicated window.

Working with Nodes

- **Add nodes:** Right-click on empty canvas space to open the categorized node menu. Type in the search bar to filter by name, class type, or category.
- **Move nodes:** Left-click and drag any node to reposition it. All selected nodes move together.
- **Select:** Left-click a node to select. Shift+click for multi-select. Click-drag in empty space for box selection. Ctrl+A to select all.
- **Delete:** Press Delete or Backspace to remove selected nodes.
- **Duplicate:** Ctrl+D to duplicate selected nodes with their connections and widget values.
- **Copy/Paste:** Ctrl+C to copy selected nodes, Ctrl+V to paste. Repeated pastes cascade with offset. Ctrl+X to cut (copy + delete).
- **Undo/Redo:** Ctrl+Z to undo, Ctrl+Y or Ctrl+Shift+Z to redo. Supports up to 50 levels of undo for all operations (add, delete, move, connect, paste, etc.).
- **Edit values:** Click widget fields on a node body to edit parameters (text, numbers, dropdowns).
- **Frame All:** Press F to zoom-to-fit all nodes in the view.
- **Auto-Layout:** Ctrl+L to automatically arrange all nodes left-to-right by dependency order.

Run-to-Node (Partial Execution)

Output nodes such as SaveImage and PreviewImage display a green **play button** in the top-right corner of their header. Clicking this button executes only the upstream subgraph required by that node, skipping any unrelated branches. This is useful for testing individual parts of a complex workflow without running the entire graph. The button has hover and press visual states for clear feedback. Use the **Cancel** button in the toolbar to stop a running generation.

Comment Groups

Group boxes are colored, titled regions that visually organize related nodes. They render behind nodes and connections with a semi-transparent fill and colored header bar.

- **Create:** Select one or more nodes, right-click, and choose "Create Group from Selection." The group wraps the selected nodes with padding.
- **Rename:** Double-click the group header, or right-click → Rename Group. A dialog appears with the current title pre-selected.
- **Move:** Drag the group header to reposition the group. All nodes inside the group move with it.
- **Resize:** Drag the small handle in the bottom-right corner of the group.
- **Color:** Right-click the group header → Color submenu. Eight preset colors are available (Blue, Green, Red, Purple, Orange, Teal, Yellow, Gray).
- **Delete:** Right-click the group header → Delete Group. This removes the group box but keeps all nodes inside it.
- **Import:** Groups from ComfyUI Web UI exports are imported automatically with their titles, positions, and colors.

Connections

Drag from an output pin to a compatible input pin to create a connection. The editor enforces type compatibility (e.g., MODEL outputs can only connect to MODEL inputs). Each input pin accepts only one connection. Double-click a wire to insert a reroute node for visual organization. Alt+click a wire to delete the connection.

Importing ComfyUI Workflows

The **Load** button accepts three formats, auto-detected on load:

- **ViewGen native format:** The editor's own JSON format with full node positions, widget values, groups, and graph state.
- **ComfyUI Web UI export:** The JSON exported from ComfyUI's web interface (File → Export). Includes node positions, groups, and widget values. Component/subgraph nodes (UUID types) are imported as placeholders.
- **ComfyUI API format:** The flat JSON format from ComfyUI's "Save (API Format)" or Developer Mode export. Nodes are auto-laid out since this format has no position data.

When importing from the Web UI format, frontend-only widget values (such as "control_after_generate" on seed inputs) are automatically detected and skipped so that widget values map correctly to node inputs.

Execution Feedback

During workflow execution, the currently active ComfyUI node is highlighted in yellow in real-time via WebSocket. This gives you visual feedback on which stage of the pipeline is running. The Cancel button in the toolbar sends an interrupt signal to ComfyUI to stop the current generation.

Results Gallery

Generated images appear in the Results panel on the right side. Click a thumbnail to view it at full size. Each thumbnail has an **X** button in the top-right corner to remove individual results. The **Clear** button next to the "Results" header removes all results at once.

7. UE Source Nodes

UE source nodes are special graph nodes unique to this plugin. They represent Unreal Engine data sources and actions that are resolved at export time, converting to standard ComfyUI nodes in the submitted workflow.

Node	Output	Description
UE Viewport Capture	IMAGE	Captures the editor viewport. Converted to a LoadImage node with the uploaded viewport PNG.
UE Depth Map	IMAGE	Extracts depth from the viewport camera. Reuses existing capture if available. Converted to LoadImage.
UE Camera Data	STRING	Injects auto-generated camera metadata (FOV, focal length, angle) into target prompt fields.
UE Prompt Adherence	—	Global override: adjusts CFG, steps, and denoise on KSampler nodes; image_fidelity and human_fidelity on Kling Image nodes; and cfg_scale on Kling Video nodes. A single adherence slider (0–1) controls all supported node types simultaneously.
UE Image Bridge	IMAGE	Saves the upstream image to disk (SaveImage) and passes it through. Creates a disk-based handoff.
UE Meshy Import	—	Downloads a Meshy Image-to-3D result and imports it as a UE asset.
UE Save 3D Model	—	Downloads a 3D model (GLB/OBJ/FBX) from ComfyUI mesh nodes and imports into UE.
UE 3D Loader	—	Browses for a local 3D model file and imports it into the Content Browser.
UE 3D Asset Export	—	Downloads generated 3D models and imports as .uasset with collision, Nanite, and placement options.
UE Image Upres	IMAGE	Upscales an image using configurable interpolation and converts between 8-bit and 16-bit colour depth.
UE Video to Image	IMAGE	Extracts a single frame from a video file via FFmpeg and outputs it as an IMAGE for downstream nodes.
UE Sequence	—	Executes downstream nodes in stages: all nodes connected to Then 0 complete before Then 1 starts.
UE GLB Extract	MESH, DIFFUSE,	Loads a GLB 3D model and outputs its geometry plus PBR texture maps as separate IMAGE pins. Browsing a GLB file

Node	Output	Description
	ORM, NORMAL	displays a thumbnail preview and texture metadata (dimensions, size) in the details panel.
UE GLB Repack	—	Takes a MESH reference and processed texture maps (DIFFUSE, ORM, NORMAL) as inputs, optionally performs UV repack via xatlas with bilinear upres (1x/2x/4x), and saves a new GLB file.

Camera Data Injection

The UE Camera Data node has configurable options: **format** (Natural Language or Structured), **position** (prepend or append to prompt), and **target** (positive only or all prompts). The plugin uses tagged markers to prevent duplicate injection across multiple generate clicks and updates stale camera data when the camera moves between captures.

8. ComfyUI Integration

Connection

ViewGen communicates with ComfyUI over HTTP and WebSocket. The server URL is configurable (default: `http://127.0.0.1:8188`). On startup or when triggered, the plugin fetches `/object_info` to discover all available node types, which populates the graph editor's Add Node menu and the ComfyNodeDatabase singleton.

Workflow Submission Flow

The generation pipeline follows these steps:

- **1. Export** — The graph editor serializes nodes to ComfyUI API-format JSON, replacing UE source nodes with standard ComfyUI equivalents.
- **2. Capture** — Viewport and depth images are captured (or reused) and uploaded via POST `/upload/image`.
- **3. Resolve** — Marker filenames (`__UE_VIEWPORT_IMAGE__`, etc.) are replaced with actual server filenames.
- **4. Inject** — Camera data is injected into target prompt fields if a Camera Data node is present.
- **5. Submit** — The workflow is sent via POST `/prompt`, returning a prompt_id for tracking.
- **6. Monitor** — WebSocket messages provide real-time node execution progress. The plugin also polls `/history/{id}` for completion.
- **7. Fetch** — The result image is downloaded via GET `/view?filename=...` and displayed in the preview panel.

Node Database

The FComfyNodeDatabase singleton caches all node definitions from the server. Each definition includes the node's class_type, category, inputs (with types, defaults, ranges, combo options), and outputs. This data drives the graph editor's Add Node menu, widget rendering, and pin type-checking for connections.

9. External API Integrations

Gemini (Nano Banana 2)

Google Gemini image generation is accessed through the NanoBanana2AIO ComfyUI node. It supports multiple model variants, aspect ratios (1:1 through 21:9), resolution tiers (1K/2K/4K), and thinking levels (MINIMAL for speed, HIGH for quality). Viewport and depth images can optionally be provided as references.

Kling Image 3.0

Kling AI image generation provides high-quality output with separate fidelity controls for general image content and human subjects. It supports model versions v1.5 through v3, various aspect ratios, and can generate up to 9 images per request. An optional viewport image serves as a reference.

Meshy (Image-to-3D)

The Meshy API converts 2D images into 3D meshes. The plugin creates an Image-to-3D task, polls for completion, downloads the resulting GLB file, and imports it into the UE Content Browser as a static mesh asset. Options include PBR texture generation and topology remeshing. Requires a Meshy API key.

10. Video Generation

The Video Generation tab provides image-to-video capabilities using a source frame (typically a generated image from the Image tab). Three backends are supported.

Backend	Key Features	Settings
Kling Video	Kling v2/v2.5 models, std/pro quality tiers	Duration 1-10s, configurable FPS, motion prompt
Wan I2V	Wan 2.1/2.6 models, 480P-1080P resolutions	Duration, FPS, prompt, negative prompt
Veo3 (Google)	Google Veo 3.0/3.1, optional audio generation	Aspect ratio, person generation policy, duration

11. 3D Model Pipeline

ViewGen includes a complete pipeline from 2D generation to 3D asset import, as well as a GLB texture pipeline for extracting, processing, and repacking textures on existing 3D models.

Image-to-3D Pipeline

- **1. Generate** an image using any generation mode.
- **2. Convert to 3D** via Meshy API (automatic task creation, polling, and download).
- **3. Import** the GLB model into UE as a static mesh with optional collision, Nanite, and auto-placement.

The graph editor provides dedicated UE source nodes for each stage: UE Image Bridge for disk handoffs, UE Save 3D Model for downloading generated meshes, UE 3D Asset Export for Content Browser import, and UE Meshy Import for direct Meshy-to-level placement.

GLB Texture Pipeline

The GLB Extract and GLB Repack node pair enables a texture processing pipeline for existing 3D models. This is particularly useful for upscaling or AI-enhancing the textures on a GLB model without modifying the geometry.

- **1. Extract** — The UE GLB Extract node loads a GLB file and splits it into separate outputs: MESH (geometry reference), DIFFUSE, ORM (occlusion/roughness/metallic), and NORMAL texture maps. Browsing a GLB file immediately displays a thumbnail preview of the first texture along with per-texture metadata (name, dimensions, file size) in the node details panel.
- **2. Process** — Connect the extracted texture IMAGE outputs to any ComfyUI processing nodes (upscale, denoise, style transfer, etc.) to enhance the textures independently.
- **3. Repack** — The UE GLB Repack node takes the processed textures and the MESH reference and rebuilds a new GLB file with the enhanced textures.

UV Repack (xatlas)

The GLB Repack node includes an optional UV repack step powered by xatlas. When enabled, the mesh UVs are unwrapped into a single optimized atlas. The pipeline bakes source textures onto the new UV layout using bilinear sampling, with configurable atlas resolution (always forced to power-of-2 for GPU performance). A dilation pass fills seam edges to prevent visible artifacts.

Upres Scale

After UV baking, an optional upres step can bilinearly upscale the baked textures by 2x or 4x. This is a post-bake operation that increases the final texture resolution beyond the atlas bake resolution, configured via the `upres_scale` dropdown on the Repack node (1x, 2x, or 4x).

12. LoRA Management

LoRA (Low-Rank Adaptation) models allow fine-tuned style control. The LoRA panel provides a dropdown of available LoRA files (auto-discovered from ComfyUI), a weight slider (0.0 to 2.0, default 0.75), and an Add button. Active LoRAs appear in a list with per-LoRA weight adjustment, enable/disable toggles, and remove buttons.

When generating, each enabled LoRA is injected as a LoraLoader node in the ComfyUI workflow, chained between the checkpoint loader and the sampling nodes. Multiple LoRAs can be stacked with independent weights.

13. Settings Reference

Connection and API settings are configured in the **Connection** tab within the ViewGen panel. Generation parameters are available in the **Basic** tab. All settings are also accessible via **Project Settings > Plugins > ViewGen** and are persisted in the project's editor configuration files.

Connection Settings

Setting	Default	Description
ComfyUI URL	http://127.0.0.1:8188	ComfyUI server address
Timeout	120 seconds	HTTP request timeout (5-600s)
ComfyUI API Key	(empty)	For paid API nodes (Nano Banana 2, partner nodes). See API Keys section for setup instructions.
Poll Interval	0.5 seconds	Status check frequency (0.25-5.0s)

Generation Parameters

Setting	Default	Range	Description
Output Width	512	64-2048	Generation output width (pixels)
Output Height	512	64-2048	Generation output height (pixels)
Denoising Strength	0.55	0.0-1.0	img2img change amount
Steps	30	1-150	Sampling iterations
CFG Scale	7.0	1.0-30.0	Prompt guidance strength
Seed	0	int64	Random seed (0 = random)
Sampler	euler_ancestral	—	Sampling algorithm
Scheduler	normal	—	Noise schedule

Depth / ControlNet Settings

Setting	Default	Description
Enable Depth ControlNet	false	Use depth map as structural guide
ControlNet Model	control_v11f1p_sd15_depth.pth	Depth control model filename
ControlNet Strength	1.0	Influence strength (0.0-2.0)
Max Depth Distance	50,000 UU	Far boundary for depth normalization (100-500,000)

Hi-Res Fix Settings

Setting	Default	Description
Enable Hi-Res Fix	false	Two-pass upscale generation
Upscale Factor	1.5	Scaling factor for second pass (1.0-4.0)
Hi-Res Denoise	0.45	Denoising for upscale pass (0.0-1.0)
Hi-Res Steps	20	Steps in upscale pass (1-150)

Cost Estimation Settings

When **Show Cost Estimates** is enabled, the status bar displays an estimated API cost each time you click Generate or Animate. Local ComfyUI modes (img2img, Depth+Prompt, Prompt Only, Wan I2V) show "Local (no API cost)". External API modes show a dollar estimate calculated from the pricing table below. Since API pricing changes over time, all values are user-editable in Project Settings.

Setting	Default	Description
Show Cost Estimates	true	Enable/disable cost display in status bar
Gemini 1K Cost	$0.067	Per image at 1K resolution (Nano Banana 2)
Gemini 2K Cost	$0.101	Per image at 2K resolution
Gemini 4K Cost	$0.151	Per image at 4K resolution
Kling Image Cost	$0.04	Per image (multiplied by Image Count)
Kling Video Std	$0.168/sec	Per second, Standard quality
Kling Video Pro	$0.224/sec	Per second, Pro quality
Veo3 Cost	$0.40/sec	Per second of generated video
Meshy Credits	15	Credits per Image-to-3D task (with texture)
Tripo Credits	20	Credits per Tripo Image-to-3D task

14. Architecture and Technical Reference

This section covers the plugin's internal architecture for developers looking to extend or contribute to ViewGen.

Module Structure

ViewGen is a single editor module that loads at PostEngineInit. It registers a nomad tab spawner via FGlobalTabmanager and extends the toolbar/menus with a launch button. The plugin includes ThirdParty integrations (xatlas for UV repacking, tinygltf for GLB parsing) alongside its core C++ source files.

Key Classes

Class	File	Purpose
FViewGenModule	ViewGen.h/.cpp	Plugin lifecycle, tab spawner, toolbar registration
SViewGenPanel	SViewGenPanel.h/.cpp	Main UI panel (SCompoundWidget)
SWorkflowGraphEditor	SWorkflowGraphEditor.h/.cpp	Interactive node graph editor
FViewportCapture	ViewportCapture.h/.cpp	Viewport RGB capture + camera metadata
FDepthPassRenderer	DepthPassRenderer.h/.cpp	Depth map extraction via SceneCaptureComponent2D
FGenAIHttpClient	GenAIHttpClient.h/.cpp	ComfyUI HTTP/WebSocket communication
FComfyNodeDatabase	ComfyNodeDatabase.h/.cpp	Singleton: node definition cache from /object_info
UGenAISettings	GenAISettings.h/.cpp	UDeveloperSettings: all configurable settings
FMeshyApiClient	MeshyApiClient.h/.cpp	Meshy Image-to-3D API client
FMeshPreviewRenderer	MeshPreviewRenderer.h/.cpp	Offscreen 3D mesh preview rendering
ViewGenGLB	ThirdParty/tinygltf/ ViewGenGLB.h/.cpp	GLB parser: load, extract textures, replace textures, save
ViewGenUVRepack	Private/ViewGenUVRepack.cpp, ThirdParty/xatlas/ ViewGenUVRepack.h	xatlas UV unwrap, texture baking, bilinear upscale

Data Flow

The generation pipeline flows through these stages: the user interacts with SViewGenPanel, which coordinates between the graph editor (SWorkflowGraphEditor), capture subsystems (FViewportCapture, FDepthPassRenderer), and the HTTP client (FGenAIHttpClient). The graph editor exports a ComfyUI-compatible JSON workflow, the panel resolves UE source node markers by uploading captured images, and the HTTP client submits the final workflow to the ComfyUI server.

GC Safety

The plugin uses a PostGarbageCollect delegate pattern to refresh stale TObjectPtr handles in heap-allocated FSlateBrush objects. Both SViewGenPanel and SWorkflowGraphEditor register for FCoreUObjectDelegates::GetPostGarbageCollect() and refresh all brush resource handles from their raw texture pointers after each GC cycle. UI textures are rooted via AddToRoot() to prevent premature collection.

Architectural Patterns

- **Singleton:** FComfyNodeDatabase and UGenAISettings provide global access.
- **Delegate/Event:** Async operations use multicast delegates (OnGenerationComplete, OnTaskProgress, etc.).
- **RAII/Unique Ownership:** TUniquePtr manages subsystem lifetimes (capture, HTTP, Meshy).
- **Weak References:** TWeakObjectPtr in FMeshPreviewRenderer prevents stale references during GC compaction.

15. Keyboard Shortcuts and Tips

Graph Editor Shortcuts

Shortcut	Action
Right-click (empty space)	Open Add Node menu with search
Right-click (on node)	Node context menu (Delete, Duplicate, Create Group)
Right-click (on group header)	Group context menu (Rename, Delete, Color)
Double-click (group header)	Rename group
Double-click (wire)	Insert reroute node
Delete / Backspace	Delete selected nodes
Ctrl+Z	Undo (up to 50 levels)
Ctrl+Y / Ctrl+Shift+Z	Redo
Ctrl+C	Copy selected nodes
Ctrl+V	Paste nodes (from internal clipboard or ComfyUI system clipboard)
Ctrl+X	Cut selected nodes
Ctrl+D	Duplicate selected nodes
Ctrl+A	Select all nodes
Ctrl+L	Auto-layout (arrange nodes by dependency)
F	Frame all nodes (zoom-to-fit)
Shift+Click	Multi-select nodes
Click-drag (empty space)	Box selection
Right-drag / Middle-drag	Pan canvas
Scroll wheel	Zoom in/out
Alt+Click (wire)	Delete connection

Tips

- **Capture once, generate many:** After clicking Capture Viewport, you can click Generate multiple times with different prompts or settings. The plugin reuses the existing capture data.
- **Import ComfyUI workflows:** Use File → Export in ComfyUI to save a workflow, then click Load in ViewGen's graph editor. Both Web UI and API formats are supported. Web UI exports preserve node positions and group boxes.
- **Run individual branches:** Click the green play button on any output node (SaveImage, PreviewImage) to execute only the upstream subgraph for that node, without running the entire workflow.
- **Organize with groups:** Select related nodes, right-click → Create Group from Selection. Use colors and titles to visually separate pipeline stages.
- **Switch between tabs** to access different workflows: Basic for quick generation, Graph Editor for custom ComfyUI pipelines, and Connection for server setup.
- **Camera data injection** is tagged to prevent duplicates. You can safely click Generate multiple times without the camera description accumulating in your prompt.
- **Gallery selection:** The currently selected result is shown at full brightness. Click any dimmed thumbnail to switch to it. The selected image also becomes the source frame for the Video Generation tab.